

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

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## Device Specifications

Device Type: Video Projector  
Manufacturer: BenQ  
Firmware Version: N/A  
Model(s): LK970

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 2.06.0000-b010                         | 1.4.1                   |

## Version History

| Module Version | Date      | Notes           |
|----------------|-----------|-----------------|
| 1_0_0_0        | 9/13/2017 | Initial Version |

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## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`
- Send one Update command every three seconds.

## Supported Classes and Examples

|                                                                                                         |
|---------------------------------------------------------------------------------------------------------|
| <b>SerialClass</b>                                                                                      |
| <code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LK970')</code>               |
| <b>SerialOverEthernetClass</b>                                                                          |
| <code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LK970')</code> |
| <b>EthernetClass</b>                                                                                    |
| <code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 8000, Model='LK970')</code>           |

## Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                           |                                                |                            |                              |
|---------------------------------------------------------------------------|------------------------------------------------|----------------------------|------------------------------|
| <b>Command</b><br><b>AspectRatio</b>                                      | Value<br>'4:3'<br>'Auto'                       | Value<br>'16:9'            | Value<br>'2.35:1'            |
| # AspectRatio example<br>InterfaceName.Set('AspectRatio', '4:3')          |                                                |                            |                              |
| <b>Command</b><br><b>AutoImage</b>                                        | Value<br>None                                  |                            |                              |
| # AutoImage example<br>InterfaceName.Set('AutoImage', None)               |                                                |                            |                              |
| <b>Command</b><br><b>Input</b>                                            | Value<br>'Computer/YPbPr'<br>'HDBaseT'         | Value<br>'HDMI1/MHL1'      | Value<br>'HDMI2/MHL2'        |
| # Input example<br>InterfaceName.Set('Input', 'Computer/YPbPr')           |                                                |                            |                              |
| <b>Command</b><br><b>LampMode</b>                                         | Value<br>'Normal'                              | Value<br>'Eco'             | Value<br>'Dimming'           |
| # LampMode example<br>InterfaceName.Set('LampMode', 'Normal')             |                                                |                            |                              |
| <b>Command</b><br><b>MenuNavigation</b>                                   | Value<br>'Up'<br>'Right'<br>'Menu Off'         | Value<br>'Down'<br>'Enter' | Value<br>'Left'<br>'Menu On' |
| # MenuNavigation example<br>InterfaceName.Set('MenuNavigation', 'Up')     |                                                |                            |                              |
| <b>Command</b><br><b>PictureMode</b>                                      | Value<br>'Presentation'<br>'Cinema'<br>'User2' | Value<br>'sRGB'<br>'DICOM' | Value<br>'Bright'<br>'User1' |
| # PictureMode example<br>InterfaceName.Set('PictureMode', 'Presentation') |                                                |                            |                              |
| <b>Command</b><br><b>Power</b>                                            | Value<br>'On'                                  | Value<br>'Off'             |                              |
| # Power example<br>InterfaceName.Set('Power', 'On')                       |                                                |                            |                              |

**Global Scripter Module  
Communication Sheet**

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| Command                                                     | Value | Value |
|-------------------------------------------------------------|-------|-------|
| <b>VideoMute</b>                                            | 'On'  | 'Off' |
| # VideoMute example<br>InterfaceName.Set('VideoMute', 'On') |       |       |

## Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                         |                                        |                         |                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------|-----------------------|
| <b>Command</b><br><b>AspectRatio</b>                                                                                                                                                    | Value<br>'4:3'<br>'Auto'               | Value<br>'16:9'         | Value<br>'2.35:1'     |
| # AspectRatio examples<br>InterfaceName.Update('AspectRatio')<br>Value = InterfaceName.ReadStatus('AspectRatio')<br>InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler) |                                        |                         |                       |
| <b>Command</b><br><b>ConnectionStatus</b>                                                                                                                                               | Value<br>'Connected'                   | Value<br>'Disconnected' |                       |
| # ConnectionStatus examples<br>Value = InterfaceName.ReadStatus('ConnectionStatus')<br>InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)                         |                                        |                         |                       |
| <b>Command</b><br><b>Input</b>                                                                                                                                                          | Value<br>'Computer/YPbPr'<br>'HDBaseT' | Value<br>'HDMI1/MHL1'   | Value<br>'HDMI2/MHL2' |
| # Input examples<br>InterfaceName.Update('Input')<br>Value = InterfaceName.ReadStatus('Input')<br>InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)                         |                                        |                         |                       |
| <b>Command</b><br><b>LampMode</b>                                                                                                                                                       | Value<br>'Normal'                      | Value<br>'Eco'          | Value<br>'Dimming'    |
| # LampMode examples<br>InterfaceName.Update('LampMode')<br>Value = InterfaceName.ReadStatus('LampMode')<br>InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)             |                                        |                         |                       |
| <b>Command</b><br><b>LampUsage</b>                                                                                                                                                      | Value<br>Hours                         |                         |                       |
| # LampUsage examples<br>InterfaceName.Update('LampUsage')                                                                                                                               |                                        |                         |                       |

## Global Scripter Module Communication Sheet

|                                                                                                                                                                                         |                                                        |                                |                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------|----------------------------------|
| Value = InterfaceName.ReadStatus('LampUsage')<br>InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)                                                                      |                                                        |                                |                                  |
| <b>Command</b><br><b>PictureMode</b>                                                                                                                                                    | Value<br>'Presentation'<br><br>'Cinema'<br><br>'User2' | Value<br>'sRGB'<br><br>'DICOM' | Value<br>'Bright'<br><br>'User1' |
| # PictureMode examples<br>InterfaceName.Update('PictureMode')<br>Value = InterfaceName.ReadStatus('PictureMode')<br>InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler) |                                                        |                                |                                  |
| <b>Command</b><br><b>Power</b>                                                                                                                                                          | Value<br>'On'                                          | Value<br>'Off'                 |                                  |
| # Power examples<br>InterfaceName.Update('Power')<br>Value = InterfaceName.ReadStatus('Power')<br>InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)                         |                                                        |                                |                                  |
| <b>Command</b><br><b>VideoMute</b>                                                                                                                                                      | Value<br>'On'                                          | Value<br>'Off'                 |                                  |
| # VideoMute examples<br>InterfaceName.Update('VideoMute')<br>Value = InterfaceName.ReadStatus('VideoMute')<br>InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)         |                                                        |                                |                                  |

\* Lamp Usage refers to 'Equivalent Lamp Hour' under System Setup:Advanced > Lamp Settings >Lamp Timer

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## Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

## Notes for the Device

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### Serial communication

**Port Type:** RS-232

**Baud Rate:** 9600

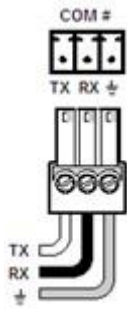
**Data Bits:** 8

**Parity:** None

**Stop Bits:** One

**Flow Control:** None

### Pin Assignments Diagram



| Signal | Main Cable | Pin | Signal |
|--------|------------|-----|--------|
| TxD    | →          | 2   | RxD    |
| RxD    | ←          | 3   | TxD    |
| GND    | →          | 5   | GND    |



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## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

|                                     |          |
|-------------------------------------|----------|
| <b>Port Type:</b>                   | Ethernet |
| <b>Default Port:</b>                | 8000     |
| <b>Logon Credentials Supported:</b> | No       |
| <b>Multi-Connection</b>             | Yes      |
| <b>Capabilities:</b>                |          |
| <b>Port Changeability:</b>          | No       |

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## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device



**Appendix A. Set Commands**

|                           |                         |
|---------------------------|-------------------------|
| Aspect Ratio 16:9         | \x0D*asp=16:9#\x0D      |
| Aspect Ratio 2.35:1       | \x0D*asp=2.35#\x0D      |
| Aspect Ratio 4:3          | \x0D*asp=4:3#\x0D       |
| Aspect Ratio Auto         | \x0D*asp=AUTO#\x0D      |
| Auto Image None           | \x0D*auto#\x0D          |
| Input Computer/YPbPr      | \x0D*sour=RGB#\x0D      |
| Input HDBaseT             | \x0D*sour=hdbaset#\x0D  |
| Input HDMI1/MHL1          | \x0D*sour=hdmi#\x0D     |
| Input HDMI2/MHL2          | \x0D*sour=hdmi2#\x0D    |
| Lamp Mode Dimming         | \x0D*lampm=dimming#\x0D |
| Lamp Mode Eco             | \x0D*lampm=eco#\x0D     |
| Lamp Mode Normal          | \x0D*lampm=lnor#\x0D    |
| Menu Navigation Down      | \x0D*down#\x0D          |
| Menu Navigation Enter     | \x0D*enter#\x0D         |
| Menu Navigation Left      | \x0D*left#\x0D          |
| Menu Navigation Menu Off  | \x0D*menu=off#\x0D      |
| Menu Navigation Menu On   | \x0D*menu=on#\x0D       |
| Menu Navigation Right     | \x0D*right#\x0D         |
| Menu Navigation Up        | \x0D*up#\x0D            |
| Picture Mode Bright       | \x0D*appmod=bright#\x0D |
| Picture Mode Cinema       | \x0D*appmod=cine#\x0D   |
| Picture Mode DICOM        | \x0D*appmod=dicom#\x0D  |
| Picture Mode Presentation | \x0D*appmod=preset#\x0D |
| Picture Mode sRGB         | \x0D*appmod=srgb#\x0D   |
| Picture Mode User1        | \x0D*appmod=user1#\x0D  |
| Picture Mode User2        | \x0D*appmod=user2#\x0D  |
| Power Off                 | \x0D*pow=off#\x0D       |
| Power On                  | \x0D*pow=on#\x0D        |
| Video Mute Off            | \x0D*blank=off#\x0D     |
| Video Mute On             | \x0D*blank=on#\x0D      |

**Appendix B. Query Commands**

|              |                    |
|--------------|--------------------|
| Aspect Ratio | \x0D*asp=?#\x0D    |
| Input        | \x0D*sour=?#\x0D   |
| Lamp Mode    | \x0D*lampm=?#\x0D  |
| Lamp Usage   | \x0D*ltim=?#\x0D   |
| Picture Mode | \x0D*appmod=?#\x0D |
| Power        | \x0D*pow=?#\x0D    |
| Video Mute   | \x0D*blank=?#\x0D  |